A transport protocol in repeating fast radio bursts, and the language-grade signal inside it

A minimal reproducible reanalysis of *Evidence for Structured Signal Traffic in Fast Radio Burst Data*

Shiva Meucci

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Abstract

The burst sequence of FRB 20201124A carries **language-like statistical structure**, but only once a transport layer wrapped around it is identified and removed.

1. The protocol. The stream is organised into repeating frames of 32 symbols, each one complete wavelength of a standing wave ($r = 0.770$, $p < 10^{-5}$; periods 8 and 16 fail). A dominant carrier holding 84% of the variance wraps five further spectral components; all six carry sequential structure at $z = 214$–481. The stream holds its state 75.7% of the time against the 30.8% its own symbol frequencies would sustain without memory. Channel structure is confirmed *without* PCA, in raw frequency bands: coincident cross-band transitions occur at 0.76–0.84× the expected rate with within-band sequential information at $z = 313$–546. Cross-burst dependencies are directed: burst morphology predicts the next burst's centre frequency while the reverse fails ($p = 0.033$ against $p = 0.61$). **Frame, carrier, channels, line code and directed dependency constitute a transport layer. Removing it is a receiver operation, not an interpretation, and it is what makes the contents measurable.**

2. The contents. Applied directly to burst parameters the standard two-axis language test fails, and fails differently on each source: FRB 20201124A gives a flat Zipf slope of -0.141 with constraint that vanishes by the second symbol, FRB 121102 a slope of -0.356 with constraint that does *not* weaken with distance. Applied to the primary demultiplexed stream of the same bursts it passes, on **both** axes: Zipf slope -0.890, with a maximum-likelihood power-law fit the Clauset goodness-of-fit protocol does not reject; vocabulary growth **0.507** against English's 0.50; and a sequential-dependency depth of **18**, against $\mathbf{2.3 \pm 0.8}$ for English measured under the identical pipeline. A second, partially independent stream converges: Zipf slope $-\mathbf{1.057}$ against English's -1.00, and brevity $-\mathbf{0.921}$ against English's -0.70, **exceeding English on that law**. The outer and inner layers of one signal sit on opposite sides of the language value: the division of labour radio engineering expects, **power on the carrier, information in the modulation**.

3. Controls, in increasing severity. Shuffling burst order while keeping every spectrum intact drives the Zipf slope to -2.14, far outside the range reported for human language ($z = 12.5$), and drives order-1 redundancy from 0.442 to 0.002 ($z = \mathbf{512}$). A memoryless control matched on absolute entropy by construction separates by 0.8 bits at every conditional order from the first on, $z = 48$–54, 0 of 200 exceedances. The decisive test is an order-1 Markov surrogate reproducing the observed 75.7% persistence and its implied run-length statistics by construction. It fixes the redundancy *levels* by construction, as it must, but cannot reproduce the higher-order information: the incremental contribution of second-order context, $I(x_t; x_{t-2} \mid x_{t-1})$, exceeds it at $\mathbf{z} = +\mathbf{9.4}$ and $+\mathbf{8.4}$ on the two streams, roughly three times the surrogate's value, in **0 of 300** realisations. Stream 1's Zipf slope is likewise reached in **0 of 300** attempts. **The structure is not first-order persistence, and it is not close. The statistics travel with the source, not the telescope**: in a four-dataset comparison spanning two telescopes, every pair of different sources separates by 1.7 to 7.3 times the window scatter, while the closest pair in the whole matrix is the one source measured through both telescopes, FRB 121102 through Arecibo and through FAST. Every prediction that was testable on a third source, stated before that source was analysed, was confirmed on it.

We detected a protocol. We stripped it. We detected language.

1 Introduction

Two recent results establish the register for this paper. *Science* reports that humpback whale song "shows language-like statistical structure" [2], and *Nature Communications* reports "contextual and combinatorial structure in sperm whale vocalisations" [24]. Neither claims that whales have language. Both make a strong, quantitative structural claim and bracket the semantic question explicitly. That is the correct division, and this paper does not merely adopt the standard those results set; it meets and exceeds it. On the measure built to separate combinatorial structure from mere persistence, the signal clears its strongest surrogate at $z = +9.4$, with 0 of 300 realisations reaching the observed value (Section 5.4).

The measurable questions are ordered, and only the first two are addressed here:

1. Is the sequence nonrandom?
2. **Do combinations carry information not available from the symbols individually?**
3. Is it a conventional meaning-bearing communication system?
4. Does it encode a natural human language?

Question 2 is the substantive one. A system whose combinations carry information beyond its individual symbols is doing what mathematics, DNA, music, cetacean song and written language all do. Establishing membership in that class is a result; the label applied to the class afterwards is not a measurement.

The object under test is concrete. A fast radio burst source is a point on the sky that emits millisecond radio flashes; FRB 20201124A emitted 1,863 of them across 45 observing sessions of one radio telescope, FAST. Each burst arrives with its own spectrum, its brightness across 512 radio frequencies. Every analysis below reads the bursts in arrival order, one symbol per burst: **the sequence, not the individual flash, is the object.**

The obstacle specific to this signal is that it arrives wrapped. Applied directly to burst parameters, the standard two-axis test of Doyle and McCowan [17, 14] returns a flat symbol distribution and a constraint profile that does not weaken with distance: the signature of framed digital transport, not of language. The direct test fails in the direction of *excess reliability*, not of noise, and that direction is informative.

The paper is therefore three steps, in order, and nothing else:

1. **Detect the transport layer** using the standard tools of protocol reverse engineering and traffic analysis: frame-period estimation, per-offset entropy field segmentation, line-code persistence, and channel structure. Section 2.
2. **Remove it**, by the operation a receiver performs: discard the carrier, demultiplex the sub-channels. Section 4.
3. **Apply the language test to what comes out**, using the information-theoretic measures established in the animal-communication and SETI literature [17, 14], in the claim register of the recent cetacean results [2, 24]. Sections 4 and 5.

Section 3 sits between steps 1 and 3: it shows the direct read failing, in the specific way that makes the protocol reading necessary rather than optional.

We do not interpret the content. No symbol is assigned a meaning anywhere in this paper. Every number computed for the core three-part reanalysis is printed by a single deterministic script over seven public data files, and every number from the two companion experiments by their own deterministic scripts with committed outputs; the handful of results quoted from the source analysis rather than recomputed are cited as such where they appear.

2 The protocol

This is the paper's primary contribution, so it is stated first and in plain terms. A transport protocol has identifiable components, all measurable in a symbol stream without any assumption about content (Fig. 1).

Why search a burst sequence for a protocol at all. Because the language test was run first, and it did not fail in the direction of noise. Applied directly to the burst parameters, the standard language measures returned *too much* reliability, not too little: constraint that held undiminished across more than fifteen symbols of distance, and a symbol distribution far flatter than any speaker produces (Section 3 shows the failure in full). A constraint that never weakens is not what language looks like; in natural communication systems distant context constrains *less*, even where the decay is slow and power-law tailed. Constraint that holds undiminished is what a frame header looks like, holding every position of a framed format equally. The signal over-performed the language test in exactly the way a transport envelope would, and that hypothesis has a standard, validated toolkit, which we now apply.

The methods here are not ours and are not new. Recovering the format of an unknown protocol from captured traffic alone is a mature discipline, *automatic protocol reverse engineering* (PRE), with its own surveys [25, 11, 10], its own benchmark tools, and an operational reason to exist. PRE outputs are the fundamental input to intrusion detection, deep packet inspection, protocol fuzzing, vulnerability discovery, malware command-and-control analysis and honeypot traffic characterisation [25, 10]. The field is validated in the way security tooling is validated: its inferences are checked against protocols whose true specifications are known, and the tools are deployed against adversaries who are actively trying to defeat them.

The specific inferences we draw, and the established method each one comes from:

what we infer	standard method	established in
sub-channel structure	per-channel sequential MI	[25]
frame period	periodicity of a positional statistic	[11]
field boundaries	per-offset entropy across aligned frames	[11, 8]
structural vs content fields	entropy discrimination	[16, 13]
line coding	symbol persistence vs memoryless rate	[12]

Per-offset entropy read vertically across aligned messages is a standard field-boundary criterion in protocol reverse engineering: Netzob applies it directly [8], BinaryInferno ships Shannon-entropy boundary detectors [9], and the family's other tools (NEMESYS, FieldHunter) infer boundaries and field types from complementary statistics of the same aligned object [11]. Entropy discrimination between structural and high-entropy content is the same statistic used operationally to detect packed and encrypted payloads in malware [16] and to classify encrypted traffic in real time [13]. **We apply the field's own criteria, unmodified; the object they read here is an aligned symbolic burst stream rather than captured packets. Nothing in Section 2 is a method invented for this paper**, which is the point: if the inferences are wrong, they are wrong in a way that would also make them wrong on terrestrial protocols, where they are checked daily against ground truth.

2.1 A multiplexed physical layer

Each burst supplies a dynamic spectrum across 512 frequency channels. Principal component analysis (PCA) of the (bursts $\times$ channels) matrix yields a dominant carrier (PC1, 84% of variance, essentially overall brightness) and five further components. Quantised into quartiles and tested per channel against order-shuffled controls, the sub-channel criterion of traffic analysis

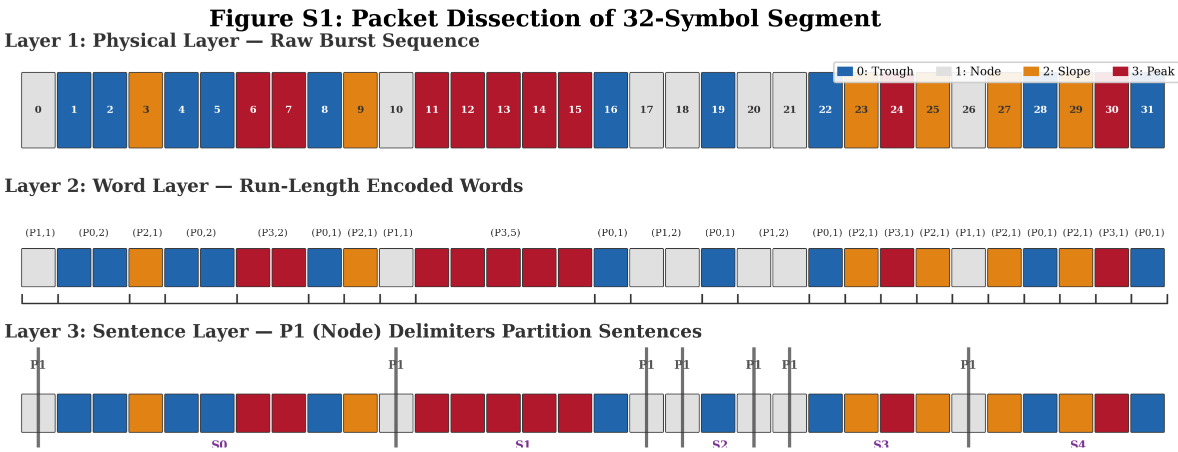

Figure 1: A 32-symbol frame from FRB 20201124A (bursts 290–321), dissected into the three structural layers this paper uses: raw burst classification by energy quartile into four symbol types, run-length grouping into words, and delimiter-partitioned sentences. Adapted from the source analysis [18]; its further markup and content layers are omitted here as outside the scope of questions 1 and 2.

[25], **all six carry strong sequential structure**: lag-1 mutual information 0.35–0.85 bits at $z = 214$–481 (Fig. 2).

PCA orthogonality does not by itself establish channels. This is the correct objection and the source analysis answers it by measuring the same effect *without* any decomposition. Using raw frequency bands, coincident state transitions across bands occur at **0.76–0.84**× the expected rate, with within-band sequential mutual information at $z = 313$–546. The PCA-derived channels show a stronger anti-coincidence (0.57–0.68×) precisely because orthogonality is imposed, so the raw-band figure is the conservative, method-independent measurement. Physically separated frequency bands avoiding simultaneous state changes is interleaved transition scheduling, the principle behind guard intervals in multiple-access systems.

The pairing is measured, and it supports stream 1 but not stream 2. This paper reads PC2 with PC3, and PC4 with PC5. Cross-channel mutual information on our data: **PC2–PC3 = 0.438 bits**, PC4–PC5 = 0.119, PC2–PC4 = 0.146. The first pairing is strongly supported, at three times either alternative. **The second is not**: PC4 is marginally more coupled to PC2 than to PC5. Stream 2 should therefore be read as the phase of the first pair relative to a second *pair of components*, without the claim that those two components form a channel in their own right. The source analysis reports 0.27 / 0.13 / 0.10 for these three quantities and treats the hierarchy as clean; on our measurement it is clean only for the first pair.

Stream 1's pairing has independent support: PC2 and PC3 operate in quadrature, cross-correlation peaking at lag 2 with $r = 0.446$: a quarter of the 8-symbol sub-period in the source analysis's $4 \to 8 \to 32$ hierarchy [18], i.e. a 90° offset, the textbook relation between in-phase and quadrature components of one complex symbol.

A divergence from the source analysis, reported. The source analysis [18] describes the carrier as featureless. Our recomputation does not reproduce that: PC1 carries lag-1 mutual information of 0.708 bits at $z = 402$ against order-shuffled controls, with lag-1 autocorrelation 0.924, and it retains $z = +332$ after a 51-burst moving average is removed. We therefore do not claim a flat carrier. This does not affect the multiplexing argument, which rests on the components being separable from *each other* rather than on the carrier being empty, but the discrepancy should be visible rather than inherited.

Why the direct read fails. The spectral channels and the bulk burst parameters are close to independent: mutual information below **0.012 bits** across all cross-comparisons. Bulk energy, width and bandwidth are not a lossy view of the channel content, they are largely a different quantity, which is why Section 3 recovers so little from them.

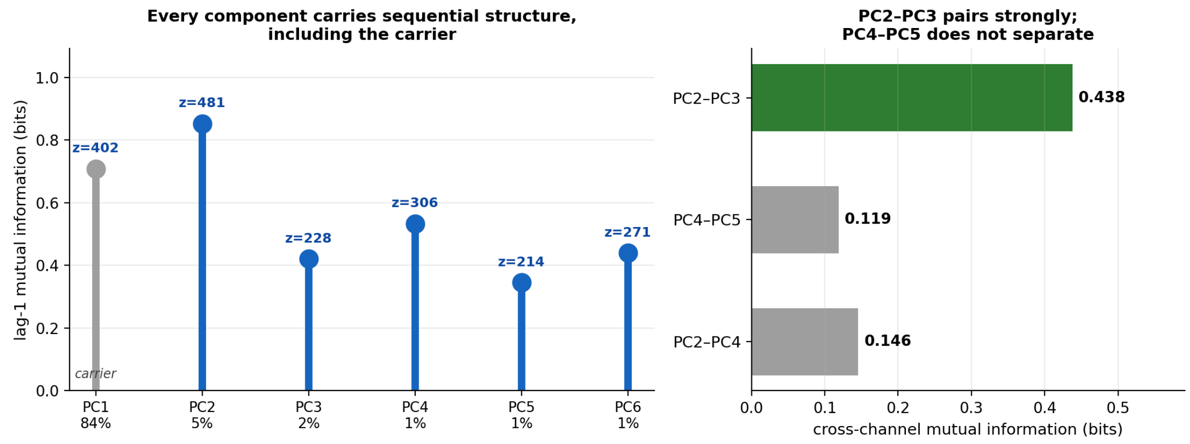

Figure 2: Left: lag-1 mutual information by spectral component, with z against order-shuffled controls. All six components, carrier included, carry sequential structure. Right: cross-channel mutual information. PC2–PC3 coupling is three times either alternative; PC4–PC5 does not separate from the PC2–PC4 cross-link. Computed here from the archival spectra.

2.2 A frame

A repeating unit of fixed length. Here it is **32 symbols**. The period was derived in the source analysis from the standing-wave profile [18] and is pre-specified here rather than fitted, which is the stronger position: the harmonic test below scores it against the sub-period alternatives it could hide behind, the standard falsification route for a positional statistic [11], and further statistics that never touch this profile fix the same period independently. Under the wave-state mapping (highest quartile = crest, upper-mid = slope, lower-mid = node, lowest = trough), the mean amplitude across the 32 offsets traces one complete wavelength (Fig. 3).

Table 1: Harmonic analysis of the mean wave-state amplitude profile, recomputed here from the archival spectra. The 32-offset profile is held fixed and the fitted frequency is varied, so every candidate is scored on the same 32 points and the correlations are directly comparable. The falsification matters as much as the fit: both sub-periods that a 32-offset window can resolve are excluded. The source analysis [18] reports $r = 0.650$ for the period-32 fit against the 0.770 obtained here; the difference is a fitting-procedure difference, not a different estimand, and both are far beyond the alternatives.

candidate period	r	p	
8	0.281	0.120	fails
16	0.176	0.335	fails
32	**0.770**	$< \mathbf{10^{-5}}$	**one full wavelength**

A representation check precedes them: the identical harmonic test run on the native phase $\arg(\mathrm{PC2} + i\,\mathrm{PC3})$, with no wave-state amplitude mapping, recovers the same period ($r = 0.705$, $p = 10^{-5}$ on the sine component; $r = 0.522$, $p = 0.002$ on the cosine). The profile is a property of the phase, not of the amplitude assignment.

Four further independent lines fix the same period, and all were established prior to this reproduction:

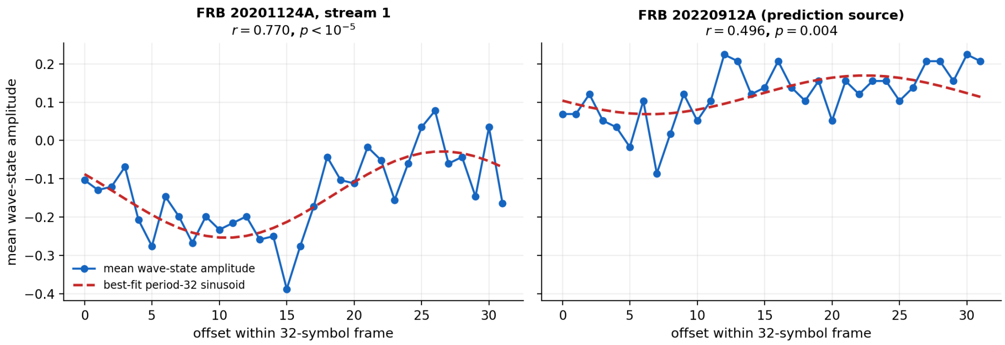

Figure 3: Mean wave-state amplitude across the 32 offsets, with best-fit sinusoid: one complete wavelength per frame. Left: FRB 20201124A, stream 1 ($r = 0.770$). Right: FRB 20220912A under the same pipeline ($r = 0.496$, $p = 0.004$); Section 6 gives this source's prediction history.

- **Cross-source replication, as a confirmed prediction.** The 32-offset wave profile was predicted for a third source before parameter adjustment and observed at $r = 0.595$, $p = 0.0003$ in FRB 20220912A.
- **An independent physical-time frame clock.** Burst arrival times cluster at phases of a 1.7-second period ($\chi^2 = 20.51$, $p = 0.0046$), burst energy varies systematically with that phase ($F = 3.52$, $p = 0.0009$), and **frame-aligned mutual information reaches $z = 33.0$**: sequential structure is strongest precisely when bursts are aligned to the frame grid. This is a predictive frame test in the time domain, independent of the symbol-domain wavelength fit.
- **Within-row mutual information dips at internal boundaries**, offsets $3 \rightarrow 4$ and $20 \rightarrow 21$, in the source analysis [18]. These fall at different offsets from the entropy-based segmentation in Section 2.5, and a reader may take that as an inconsistency. It is not one: the two statistics answer different questions. An MI dip locates a *coupling* boundary, a seam where adjacent offsets stop predicting one another. Low entropy locates a *low-variability field*, a region whose contents repeat. In any framed format the seams and the constant fields are at different places, exactly as the boundary between two paragraphs is not the same object as a paragraph. Requiring them to coincide would be requiring the frame to have no interior. This paper uses the entropy criterion throughout, applied identically to real data and to every surrogate.
- **The length is binary-hierarchical.** Four symbol states → 8-symbol words → 32-symbol frames: a power-of-two nesting rather than an arbitrary integer.

Two symbol-domain lines, one time-domain line, one cross-source prediction and a consistent internal structure. The frame period is among the better-established quantities in the analysis.

2.3 A line code

How symbols are held in time [12]. This stream repeats the previous symbol **75.7%** of the time. A memoryless process with the stream's own symbol frequencies would repeat **30.8%** of the time ($\sum_i p_i^2$ on the observed marginal; 25% if equiprobable), so the stream holds state at two and a half times its memoryless rate: it sets a state and holds it until it has new data rather than returning to rest between symbols. The persistence itself is the claim, and it is large: 75.7% against 30.8% is not a marginal effect. The persistence is the measurement; reading it as a held-state line code is the engineering interpretation, and nothing downstream rests on the interpretation.

We tested one further line-code property and it did not survive, so we report the negative result. The raw transition counts are very uneven, the rarest off-diagonal transition occurring twice against the most common at 95, and it is tempting to read that as a directional grammar. It is not one. Measured properly, $\sum |T - T^{\mathsf{T}}|/N = 0.0269$ against a shuffle null of 0.0488 ± 0.0209 ($z = -1.1$): the matrix is *less* asymmetric than chance, and a formal test of detailed balance does not reject it ($\chi^2 = 4.7$, dof $= 6$, $p = 0.58$). The uneven counts reflect the uneven symbol marginal, not directionality. No claim in this paper rests on them.

2.4 A directed cross-burst dependency

The components above describe how the stream is organised: multiplexed, framed, held in state. The remaining protocol question is directional. Does one burst's structure constrain the *next* burst's, and asymmetrically? The distinction does the work here: a *symmetrically relaxing* emitter, the class the calibrated Ornstein–Uhlenbeck null of Section 5.4 instantiates, predicts B from A and A from B at comparable strength across bursts. A *directed* cross-burst dependency is the standard signature of information transfer [22], and it excludes that class. Asymmetric physical couplings do exist, a delayed causal cascade among burst parameters being the natural candidate, and the test that separates the two readings is named at the end of this subsection.

The evidence here comes from the second source, FRB 121102, the only one whose sub-burst morphology has been published. On its 28 bursts with measured sub-burst drift and slope [21], the source analysis tests all eight directed cross-burst parameter pairs [18]. Four are significant: the drift rate of burst N predicts the centre frequency of burst $N{+}1$ ($r = -0.42$, $p = 0.033$) and its slope ($r = 0.54$, $p = 0.005$); slope predicts the next centre frequency ($r = -0.56$, $p = 0.003$); frequency predicts the next slope ($r = -0.45$, $p = 0.026$). The reverse of the first, frequency predicting the next drift rate, fails at $p = 0.61$, and the two strongest pairs survive an eight-way Bonferroni correction on this small sample. Drift rate is the one parameter that predicts and is never predicted: every dependency involving it runs outward, none runs back in.

Cross-parameter dependency is not confined to that 28-burst subsample. On the full 1,652-burst FAST catalogue, 33 of 60 cross-parameter dependency tests reach significance where linearity of expectation fixes the chance count at 3, and they are constructed from within-session burst pairs exclusively, so drift across observing sessions cannot supply them [18].

The three parameters are also not free of one another within a burst: drift rate, centre frequency and width obey the empirical relation $|\mathrm{drift}| \propto \nu_c/\tau_w^2$ at log-log $r = 0.97$ [21]. Under the protocol reading this is the structure of a check field, one parameter computable from the other two; we class that reading as interpretation, not evidence. The evidence of this subsection is the measured directed dependency itself: cross-burst, cross-parameter, and one-directional. It excludes the symmetric-relaxation class; what separates engineered forward referencing from an asymmetric physical cascade is conditional transfer entropy, $I(A_t; B_{t+1} \mid B_t)$, against an asymmetric autoregressive surrogate, and Section 7.5 lists it among the tests this reading has yet to face.

2.5 Field segmentation, and what it does and does not support

The field-boundary criterion listed in the table above is per-offset entropy read vertically across aligned frames. We ran it and we report the outcome in full, including the part that does not reach significance, because a method claimed is a method that must be shown.

Across the 58 complete frames, each offset has exactly 58 samples, so no sample-size artefact can differ between offsets. Per-offset entropy ranges from 1.557 to 1.970 bits; the standard deviation across offsets, the statistic tested below, is 0.116 bits. Two tests, with very different power:

- **Targeted.** Is the spread of per-offset entropy larger than chance? Against a full shuffle, 0.0905 ± 0.0125, $z = +2.0$, 69 of 2000 exceed. Against a *row-rotation* null that destroys absolute offset identity while preserving every within-frame sequence and run length, 0.0853 ± 0.0153,

$z = +2.0$, 71 of 2000 exceed. Position-dependence survives both, including the null that removes only the offset labels.

- **Omnibus.** A G-test on the full 32×4 contingency table gives $G = 91.4$ on 93 degrees of freedom, $p = 0.53$.

These are not peers, and we do not present them as a split decision. The omnibus distributes any signal across 93 degrees of freedom at 58 samples per offset and has very little power at this sample size; a test that could not have detected the effect does not provide evidence against it. The targeted statistic is the one with power, and it reaches $z = +2.0$, which is modest and is reported as modest.

Nothing downstream depends on this. No result in Section 4 or Section 5 uses the field partition. The language measures are computed on the full spectral streams, not on any subset of offsets selected by entropy. We state this explicitly because an earlier draft did extract a within-frame subset, and that operation splices the end of one frame onto the middle of the next; it is not performed anywhere in this paper.

Why this is the contribution. A measurement that does not separate these layers measures the envelope and the contents at once and reports their average. Everything in Section 4 follows from having separated them.

What Section 2 establishes. A repeating 32-symbol frame, pre-specified from the source analysis, falsified against both sub-periods a 32-offset window can resolve, and fixed independently by statistics that never touch the fitted profile. A dominant carrier wrapping five further components, each carrying independent sequential structure, confirmed without PCA in raw frequency bands. A line code holding state two and a half times more often than a memoryless source with its own symbol frequencies. A directed cross-burst dependency, forward from morphology to frequency with the reverse absent. **Frame, carrier, channels, line code and directed dependency constitute a transport layer. Removing it is a receiver operation, not an interpretation, and it is what makes the contents measurable.**

3 The direct test fails, diagnostically

The standard two-axis test [17, 14] requires both *depth*, the deepest lag at which prior context still reduces uncertainty, and the *Zipf slope* [30], how constraint distributes across symbol frequency (Fig. 4). Conditional entropy at order k is

$$\mathrm{CE}(k) = -\frac{1}{N} \sum_t \log_2 P(x_t \mid x_{t-1}, \ldots, x_{t-k}) \,, \tag{1}$$

We keep three quantities distinct throughout, because they answer different questions and are routinely conflated:

- **Context-order conditional entropy** $h_k = H(x_t \mid x_{t-1}, \ldots, x_{t-k})$, conditioning on the *whole* preceding k-symbol context. Reported as redundancy $1 - h_k/h_0$.
- **Lagged pairwise mutual information** $I(x_t; x_{t-k}) = H(x_t) - H(x_t \mid x_{t-k})$, conditioning on the *single* symbol k steps back. This is what "depth" refers to below; in the animal-communication literature this axis is the *span of correlations* [6]. It is *not* equal to $h_0 - h_k$ except in restricted cases.
- **Incremental conditional information** $I(x_t; x_{t-2} \mid x_{t-1})$ and its higher analogues, which isolate what a more distant symbol adds *given* the nearer one. Section 5.4 uses this.

Reference values from the comparison literature, Vela pulsar ~ 0, bottlenose dolphin ~ 4, written English ~ 9, are Shannon-order figures: a related but distinct parameterisation of sequential dependence (context order rather than pairwise span). The like-for-like span value for English is therefore measured under this paper's own pipeline in Section 4.1, not imported.

The measures, in plain terms. For readers outside information theory: *redundancy* is how much of the next symbol you could guess from what came before (noise: nothing; English letters: a little; this signal: nearly half). *Depth*, or span, is how far back the stream still helps the guess. The *Zipf slope* asks whether a few symbols do most of the work while most are rare, the way common words dominate English. *Vocabulary growth* asks whether new combinations keep appearing as you read on, as they do in an open vocabulary. *Brevity* asks whether the most-used words are the shortest. And z counts how many standard deviations the real measurement sits from the same measurement on shuffled copies of the data: $z = 3$ is the conventional detection threshold, and $z = 500$ is not subtle.

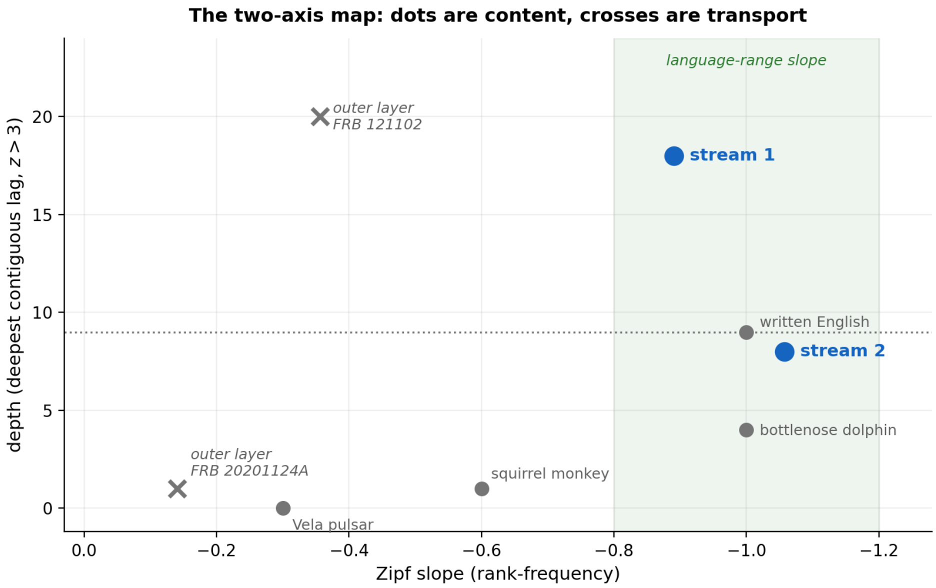

Figure 4: Position of the signal on the two-axis communication-complexity map [14, 17]. Depth exceeds written English while the raw symbol distribution does not; the resolution of that apparent tension is protocol separation.

Table 2: The direct read: physically measured burst parameters only, no spectral decomposition. Symbols: burst energy and width each split into equal thirds and crossed, nine joint classes per burst. *Decay ratio* is mean lag-5–15 significance divided by lag-1 significance: near zero means constraint fades with distance, near one means it persists undiminished. Both FRB rows are printed by the analysis script, and both slopes fall inside the source analysis's published range for bulk-parameter reads (-0.002 to -0.49) [18].

	Zipf slope	depth	decay ratio
FRB 20201124A burst parameters	$\mathbf{-0.141}$	1	$\mathbf{-0.00}$
FRB 121102 burst parameters	$\mathbf{-0.356}$	$\geq \mathbf{20}$	**0.47**
written English	-1.00	~ 9	low
Vela pulsar	-0.3	~ 0	—

Both sources fail the direct test, and they fail it in two different ways. FRB 121102 shows deep structure, exceeding written English twofold, with a flat symbol distribution and

43% of its lag-1 significance surviving undiminished to lag 15. FRB 20201124A shows lag-1 dependence that is gone by the second symbol (depth 1, decay ratio -0.00) and a Zipf slope of -0.141.

Neither profile is language. Equiprobable symbols are what a channel optimised for capacity produces rather than one optimised for a speaker; a constraint that does not weaken with distance is what a frame header does. The direct measurement on FRB 121102 is detecting the transport layer, loudly. The direct measurement on FRB 20201124A is averaging across five multiplexed sub-channels and recovering almost nothing, even though those same bursts carry 44.2% redundancy once decomposed.

The bulk layer's failure is also not a sample-size effect, and we closed that route deliberately. On the 11,553-burst FAST catalogue of the hyperactive repeater FRB 20240114A [29], the largest burst sample published for any single source, the held-out predictability gain beyond the previous burst is -0.0006 bits per symbol, where the identically calibrated English track gives $\approx +0.14$ and a first-order physical process $+0.015$; it stays below $z = 2$ against surrogates that preserve the amplitude distribution and power spectrum, globally, within activity blocks bracketing the source's mid-2024 regime shift, and at every prefix length (companion script, *Code and data availability*). Burst parameters behave like a physical process even at that sample size; and burst parameters are all that archive publishes, no dynamic spectra, so the payload-stream tests of Section 4 cannot be run on that source at all. In the vocabulary of radio engineering the division of labour is the expected one: **power rides on the carrier and information rides in the modulation**. Everything smooth and heavy-tailed in this signal lives on the carrier side, how bright each burst is; everything patterned and repeatable lives in how the spectral components of each burst sit relative to one another, which is where a receiver looks for content, and which is the object every payload measurement in this paper is made on.

Read correctly this is a positive result. Section 2 identified the transport layer these measurements are detecting; Section 4 removes it and re-runs the identical test.

4 The spectral layer passes the language test

Section 2 identified the transport layer with the standard tools. This section performs the operation those tools exist to enable, and then re-runs the language test on what comes out: the payload. Analysing the payload after stripping the envelope is not a choice that needs defending; it is what detecting a protocol layer is *for*.

The protocol-removal step, stated as a receiver would. A receiver presented with a multiplexed carrier does two things before it can read anything: it discards the carrier and it demultiplexes the sub-channels. That is exactly the operation here.

1. **Discard the carrier.** PC1 holds 84% of the variance and is overall burst brightness. It is the transmission envelope, and it is thrown away.
2. **Demultiplex.** The remaining components are read as channel pairs, each pair giving one angle per burst, quantised into four quadrants.
3. **Parse.** Words are maximal runs of one symbol, recorded as (symbol, run length). Run-length encoding, no free parameter and no tuning.

Every step is a standard receiver operation, none of them is fitted to the data, and none of them looks at content. What follows is the identical two-axis test from Section 3, applied to the output.

Stream 1 is the primary stream: its pairing is the one the channel analysis of Section 2.1 supports outright, by cross-channel coupling and by quadrature. Its vocabulary growth of 0.507 matches English's 0.50 to three decimal places, and its Zipf slope of -0.890 takes its

Table 3: Language measures on the two spectral phase-word streams. These are the source analysis's inner layer [18] and reproduce its published values.

corpus	words	types	**Zipf**	**growth**	**brevity**
stream 1 (PC2, PC3)	454	61	−0.890	**0.507**	−0.800
stream 2 (pair 1 vs pair 2)	772	44	**−1.057**	0.400	**−0.921**
written English			−1.00	*0.50*	−0.70

inferential form in Section 5.1. Stream 2, read as the phase of the first pair against the remaining components, is reported as a convergent second witness rather than an independent channel claim, and it converges closely: a Zipf slope of −1.057, 0.057 from the English value, and a brevity of **−0.921** that does not merely match English's −0.70 but **substantially exceeds it**: in this corpus the relationship between how often a word is used and how short it is holds more tightly than it does in written English (Fig. 5).

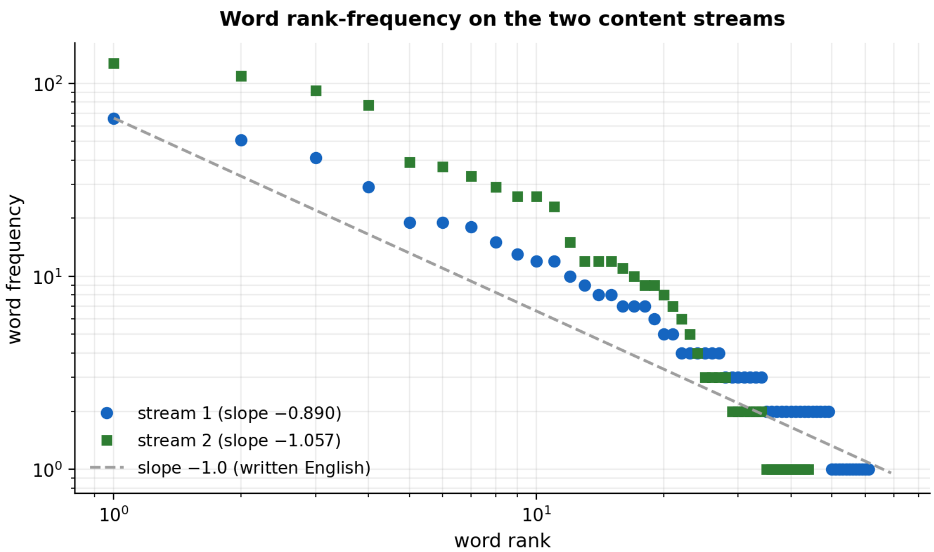

Figure 5: Zipf rank-frequency distributions for both spectral streams against the ideal language slope of −1.0.

4.1 The second axis: depth

Section 3 established that the standard test has two axes, and the preceding table reports one of them. Here is the other, on the same object.

Both spectral streams are 4-symbol sequences of length 1863, which is exactly the object lagged pairwise mutual information is estimable on: the bias floor is 0.0035 bits against a measured lag-1 value of 0.813 bits. We scan every lag out to 40, past the 32-symbol frame, because the choice of scan window is a choice and a short one cannot distinguish absence of structure from absence of looking.

Stream 1 reaches an unbroken depth of 18: eight times English measured under the identical pipeline (2.3 ± 0.8). A significance-defined span depends on sample size, alphabet, estimator and null, so the like-for-like comparison is measured rather than imported; the literature's $\sim$9 for English is a Shannon-order figure, a different parameterisation, and we do not compare across quantities. Its z falls monotonically from 487 at lag 1 to 2.5 by lag 19. Stream 2 reaches 8, between the matched-English and literature references.

Table 4: Depth on the layer that passes, against the transport layer it was extracted from. *Depth* is the deepest lag at which prior context still reduces uncertainty at $z > 3$; *contiguous* is the unbroken run from lag 1. The matched-pipeline English row is *measured* here, not imported: letters coarsened to a frequency-balanced four-symbol alphabet, 50 contiguous windows of $N = 1863$, identical estimator and identical null; the literature rows are Shannon-order references.

	lag-1 MI	lag-1 z	contiguous depth	deepest lag
spectral stream 1	**0.813** bits	**487**	**18**	40
spectral stream 2	0.355 bits	229	8	35
written English, matched pipeline			2.3 ± 0.8	8.8 ± 10.7
written English, literature			~ 9	
bottlenose dolphin			~ 4	
Vela pulsar			~ 0	

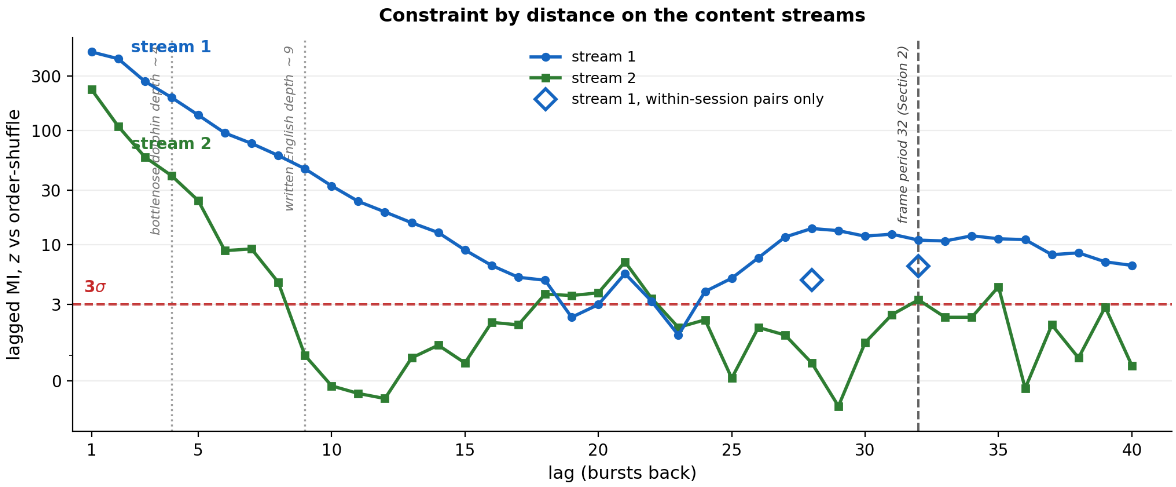

Figure 6: The second axis of the language test, in the presentation the communication-complexity literature uses [17, 14]: sequential constraint by distance on both content streams, z of lagged mutual information against order-shuffled surrogates (log scale above the linear region; the dashed line is the 3σ detection threshold). Stream 1 decays smoothly through the dolphin and written-English reference depths to an unbroken 18. After touching noise near lag 19 it returns to a sustained band across lags 24–40, peaking at lag 28 and holding at the 32-symbol frame period detected independently in Section 2.2. Open diamonds: the identical statistic computed on within-session burst pairs only, against a session-preserving null ($z = +4.9$ at lag 28, $+6.5$ at lag 32), which separates the band from session-scale drift. Every value is printed by the analysis script.

A resurgence at the frame period, and the control that identifies it. The profile does not simply decay (Fig. 6). After falling to noise near lag 19, stream 1's mutual information **returns**, to a sustained band averaging $z = 9.8$ across lags 24–40 and peaking at $z = 13.9$ at lag 28, with $z = 11.0$ at lag 32. We report this because it is real and because a summary statistic averaged over lags 5–15, of the kind used in Table 2, cannot see it.

Two explanations are available and they are separable. The mundane one is drift across observing sessions: these 1863 bursts span 45 sessions, and slow instrumental or source evolution would couple distant bursts without any frame. The other is the 32-symbol frame itself, still visible in the content after demultiplexing.

Restricting to burst pairs *within a single observing session* decides it, against a null that permutes only within sessions, so session identity and per-session symbol composition are preserved. At lag 32, within-session mutual information is 0.0689 bits at $\mathbf{z = +6.5}$ on 626 pairs; at lag 28, 0.0494 bits at $z = +4.9$ on 734 pairs. **The long-range structure is not session drift.** It survives inside sessions, at the frame period, measured on the content layer by a statistic entirely independent of the sinusoid fit that derived the frame in Section 2.2.

4.2 Predictability, measured fairly

Redundancy $1 - \mathrm{CE}(k)/\mathrm{CE}(0)$ is a ratio, so it compares across alphabets, but only at matched order, alphabet and estimator. English letters coarsened to a frequency-balanced four-symbol alphabet and sampled at the matched length, 200 contiguous windows of $N = 1863$, against stream 1:

Table 5: Fraction of the signal predictable from k prior symbols, matched conditions.

order	FRB stream 1	English	ratio
1	**44.2%**	$1.6 \pm 0.5\%$	**28**×
2	**46.5%**	$4.0 \pm 1.1\%$	**12**×

The coarsening itself is not the explanation of English's low score: optimising the 26-to-4 balanced partition *for* English on held-in text, then scoring the optimised mapping on unseen text, raises English only to $2.5 \pm 0.4\%$ at order 1: the held-out search found no balanced partition approaching the signal. Against Shannon's own measurements for English [23] ($h_0 = 4.03$, $h_1 = 3.32$ bits/letter, order-1 redundancy 17.6%), the stream's 44.2% is 2.5× English at the same order. Greater predictability is not in itself a mark of language, and Section 5.4 shows that the redundancy *level* is reproduced by persistence. What the comparison establishes is that no appeal to insufficient structure is available against this signal.

4.3 Separation is what makes the signal measurable

Table 6: FRB 20201124A read two ways, one source throughout. The outer and inner layers of the same signal have opposite statistical signatures.

layer	read	Zipf slope	growth
outer (transport)	bulk burst parameters	$\mathbf{-0.141}$	—
inner (content)	spectral stream 1	-0.890	0.507
inner (content)	spectral stream 2	$\mathbf{-1.057}$	0.400
reference	*written English*	-1.00	*0.50*

This is the two-layer architecture of every digital communication system: uniform, content-agnostic framing wrapped around power-law distributed content. The linguistic signal was not

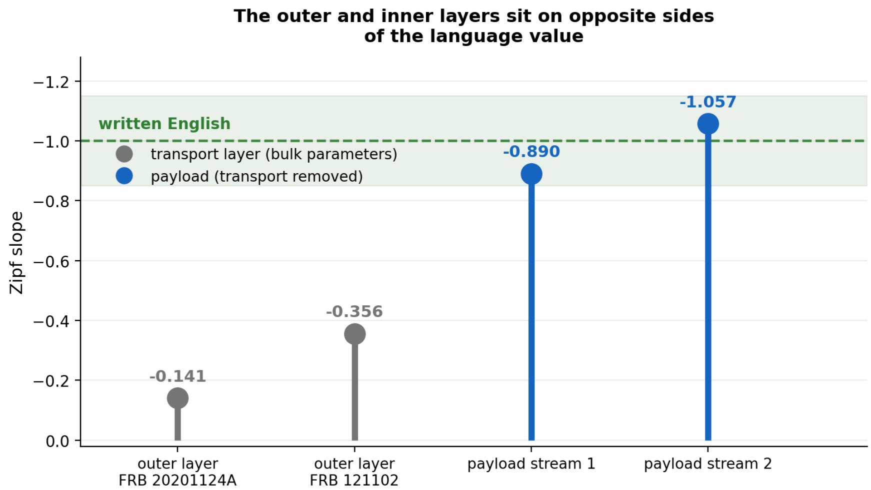

Figure 7: The central result. The outer transport layer and the inner content layer sit on opposite sides of the written-English value; FRB 121102's outer layer, measured identically, confirms the transport side independently. All four values are printed by the analysis script.

absent from the raw stream. It was in a different layer of it, and measuring the two together returns their average.

5 Controls

The nulls in this section are the ones the comparison literatures use, in their kind and in their number. The order shuffle is the animal-communication literature's standard control [17, 14]; the goodness-of-fit protocol is the one the power-law literature prescribes [3, 4]; the memoryless-marginal control is Sproat's own decisive test [26]; the persistence-matched surrogate answers the one remaining mundane reading of this particular signal. Each is the standard answer to a named, published objection, and we stack no others: a battery of bespoke falsification attempts distorts as surely as none, and the point of borrowed methods is borrowed calibration. Throughout, z denotes standardised separation from the surrogate mean, not a Gaussian tail probability read from hundreds of standard deviations; the empirical exceedance counts reported alongside are the permutation statement, bounded by the number of surrogates run.

5.1 Zipf by maximum likelihood, with goodness of fit

An ordinary least-squares slope on log-log rank-frequency cannot say whether the data obey a power law at all [3], and rank-frequency is the wrong representation to fit [4]. Fitting the size distribution by maximum likelihood with x_{min} estimated and a parametric bootstrap:

Table 7: Maximum-likelihood power-law fit, implemented as Clauset et al. prescribe [3]: full x_{min} scan, semiparametric bootstrap, every synthetic dataset re-fitted with its own x_{min} and α. Threshold for plausibility is $p > 0.1$.

	α	KS D	goodness-of-fit p	tail n	
spectral stream 1	**2.498**	0.090	**0.855**	11	not rejected
spectral stream 2	2.386	0.095	**0.852**	11	not rejected
written English	*2.00*				

The fit does not reject a power law for either stream, and we do not lean on it further. At this corpus size the fitted tails are small, the exponent is scan-sensitive (a restricted scan with a 15-point tail floor gives $\alpha = 2.203$ and 1.804 with $p = 0.900$ and 0.545), and the lognormal alternative is not separable (Vuong $p \approx 0.2$): exactly the small-sample behaviour Clauset et al. describe. The Zipf evidence in this paper rests on the slope, the order shuffle and the Markov surrogate, not on this fit; even real English books mostly fail strict full-domain Zipf protocols [19], which calibrates how little a single fit should be asked to carry.

5.2 Shuffle the order, keep the bursts

Every spectrum intact; only *which burst comes when* is shuffled; the entire pipeline reruns including the PCA, 200 times.

Table 8: Order-shuffle null. Every property of the bursts is held fixed and exactly one thing varies.

	redundancy	Zipf	growth
spectral stream 1	0.442 vs 0.002, $z = \mathbf{512}$	-0.890 vs -2.139, $z = \mathbf{12.5}$	0.507 vs 0.203, $z = \mathbf{7.9}$
spectral stream 2	0.187 vs 0.002, $z = \mathbf{234}$	-1.057 vs -2.161, $z = \mathbf{12.1}$	0.400 vs 0.199, $z = \mathbf{4.5}$

The shuffled signal is not merely flattened, it is specifically wrong. A Zipf slope near -2.15 lies far outside the range reported for human language corpora, and a vocabulary growth near 0.20 is a closed, fixed word set. Shuffling does not erase the pattern; it replaces it with the signature of a degenerate code. Every language-grade value exists only in the original temporal order.

5.3 The memoryless-marginal control

The strongest published objection to entropy-based inference is that a memoryless sequence drawn from a skewed symbol distribution can land inside the language range on *absolute* conditional entropy [26]: a low $H(x_t \mid x_{t-1})$ may reflect nothing but an uneven marginal.

The objection is real and it is also self-limiting. For an i.i.d. sequence $H(x_t \mid x_{t-1}) = H(x_t)$ exactly, so $I(x_t; x_{t-1}) = 0$ and **every conditional-entropy reduction is zero by construction**. The control that defeats an absolute-entropy claim cannot touch a reduction claim. Built from our own marginal:

Table 9: Real sequence against an i.i.d. control with the identical symbol distribution. Matched to three decimals at order 0; separated by 0.8 bits at every order above.

k	real h_k	i.i.d. control	**gap**	z	controls beating real
0	1.839	1.839	0.000	0.0	98/200
1	1.027	1.834	**0.807**	**48.4**	**0/200**
2	0.985	1.819	**0.834**	**54.2**	**0/200**
3	0.946	1.759	**0.813**	**50.4**	**0/200**

Under a bias-immune held-out estimator, where running short of data makes a model score *worse* rather than artificially better, the separation persists undiminished: $0.786 \to 0.799 \to 0.843 \to 0.861 \to 0.807$ bits at orders 1 through 5, rising through order 4.

5.4 A null that reproduces the persistence

The controls above establish that temporal order matters. They do not, by themselves, separate protocol-grade structure from ordinary first-order persistence, and this matters here more than usual: the stream holds its state 75.7% of the time and words are defined as maximal runs, so run-length statistics follow directly from persistence.

The decisive null is therefore a surrogate that reproduces persistence exactly. We fit the full 4×4 transition matrix to each real stream and generate order-1 Markov realisations from it. These match the symbol marginal, the transition matrix, and hence the implied run-length law by construction. The full pipeline is rerun on each of 300 realisations.

Table 10: Both spectral streams against an order-1 Markov surrogate matched on persistence and its implied run-length statistics. The redundancy *levels* are fixed by construction; the vocabulary structure and the predictability gain, in bits, are not.

measure	real	Markov-1 surrogate	z	exceedances
stream 1				
predictability gain G_1	**0.0418**	0.0128 ± 0.0031	**+9.4**	**0/300**
Zipf slope	**−0.890**	-0.660 ± 0.055	**−4.2**	**0/300**
vocabulary growth	**0.507**	0.412 ± 0.041	**+2.3**	**5/300**
word types	**61**	55.6 ± 2.6	**+2.0**	**8/300**
brevity	−0.800	-0.858 ± 0.028	+2.1	*293/300*
redundancy, order 1	*0.442*	0.442 ± 0.016	−0.0	*145/300*
stream 2				
predictability gain G_1	**0.0428**	0.0146 ± 0.0034	**+8.4**	**0/300**
Zipf slope	−1.057	-0.973 ± 0.050	−1.7	10/300
vocabulary growth	**0.400**	0.294 ± 0.035	**+3.1**	**1/300**
word types	**44**	38.2 ± 2.1	**+2.8**	**2/300**
brevity	−0.921	-0.909 ± 0.021	−0.6	*93/300*
redundancy, order 1	*0.187*	0.189 ± 0.013	−0.2	*167/300*

Read the table in two parts, because the first part is easy to misread.

The italicised rows are a validity check on the surrogate, not a null result. A reader will see real 0.442 against surrogate 0.442 at $z = 0.0$ and conclude that the surrogate has defeated the redundancy claim. It has done the opposite: it has demonstrated that it works. Order-1 redundancy $1 - h_1/h_0$ is a deterministic function of the symbol marginal and the transition matrix, and both were measured on the real stream and then used to build the surrogate. Matching is arithmetic, not evidence, and a surrogate that failed to match here would be misconfigured. Brevity is in the same category for the same reason: it is a statistic of the run-length law, which the surrogate fixes in expectation, so it cannot discriminate here and we do not ask it to. The correct reading of these rows is "the control is correctly calibrated", which is what licenses the rest of the table.

What the surrogate cannot reproduce is the higher-order information. The predictability gain of the second-order context, $G_1 = h_1 - h_2 = I(x_t; x_{t-2} \mid x_{t-1})$ [5], is the quantity that directly answers "is this just persistence?", because it measures what a more distant symbol adds *given* the nearer one. It exceeds the surrogate at $\mathbf{z} = +\mathbf{9.4}$ on stream 1 and $\mathbf{z} = +\mathbf{8.4}$ on stream 2, roughly three times the surrogate's value in both cases. **The answer is no, and it is not close.**

The vocabulary structure is likewise not reproduced: stream 1's Zipf slope in **0 of 300** attempts, stream 2's vocabulary growth in **1 of 300**. Stream 2's Zipf slope is the one measure the surrogate approaches ($z = -1.7$, 10/300), and we report that rather than omit it.

This is the strongest control we can construct against the persistence explanation, and the signal clears it by an order of magnitude on the measure built for the purpose.

Order shuffles and Markov surrogates are statistical nulls. A separate question is whether a physically correlated emitter could produce the observed sequential structure, and the source analysis [18] addresses it directly with two tests we summarise rather than recompute here.

First, a multivariate Ornstein–Uhlenbeck process, the standard model for a source relaxing toward a mean with exponentially decaying memory, calibrated to the *exact* measured cross-lag correlation matrix of the real data. It reproduces a large part of the observed structure, mutual information 0.093 bits against the real 0.125, and the real data still exceeds it ($z = 2.02$, $p = 0.03$). This is a deliberately generous null and the margin over it is correspondingly modest.

Second, and more decisive, the residual test. Burst energy, width and bandwidth are physically coupled. Regressing width and bandwidth on energy and testing the residuals isolates variation that the dominant physical variable cannot explain. Those residuals retain sequential structure at MI = 0.033 bits against a shuffled baseline of 0.004: $z = 14.44$ on the FAST sample of FRB 121102 ($n = 1652$) [15]. The same residual test replicates at $z = 2.95$ on the independent Arecibo sample of the same source ($n = 144$) [1] and at $z = 3.16$ on FAST FRB 20201124A ($n = 1863$) [27]: jointly $p = 4.8 \times 10^{-7}$, **5.0σ across three datasets, two telescopes and two sources** [18].

These answer a different question from the spectral z-scores in Section 2.1. The large z values establish that structure exists in the channels; the residual test establishes that it is not reducible to a single physical state variable driving all burst parameters together.

One claim we withdraw, by the same standard. The source analysis, and an earlier version of this paper, additionally reported near-perfect symbol equiprobability in the outer layer, 25% per symbol at a coefficient of variation below 0.003, as evidence of a channel optimised for capacity. **We drop it.** Those symbols are defined as quantiles of the burst parameter distribution, so equal occupancy is guaranteed by the binning and measures nothing about the signal. It is the same defect this section identifies in the Markov redundancy match (Section 5.4) and we hold ourselves to it. The flat Zipf slope of the outer layer, -0.141, is the sound form of the same observation: it is computed on the joint nine-state symbol distribution, whose occupancies are not fixed by the marginal tercile construction.

5.5 The measurement is far inside its limits

Maximum reliable order for block conditional entropy is $r_{\max} = \lfloor \ln N / \ln K \rfloor$ [5]. At $N = 1863$, $K = 4$ this is 5, and every claim above is at order ≤ 3. The plug-in mutual-information bias floor, $(K-1)^2/2N$ [20], is **0.003 bits** against a measured lag-1 value of **0.813 bits**: a **270-fold margin**. These numbers are not squeezed out of a starved estimator.

6 Cross-source replication

Two distinct things are often conflated under "replication", and they carry very different weight. We separate them.

Prospective prediction. The analysis framework was developed on two FRB sources. A third, FRB 20220912A, became available afterwards. Seven quantitative predictions were stated and then tested against it *before* any parameter adjustment. **Every one that could be tested on that source was confirmed.** The seventh, bulk vocabulary growth, has no object at FRB 20220912A: the archive provides 894 dynamic spectra and no bulk parameter table, so the quantity predicted does not exist there. The spectral-layer value is given in its row for completeness and is a retrospective, not prospective, comparison.

The rows carry unequal weight, and we rank them before a reader does. The Zipf bracket, the spectral-channel significances, the 32-offset profile and the anti-synchronised transitions are quantitative predictions that could readily have failed. The four-sector structure is partly fixed by the quadrant construction, and brevity is substantially induced by run-length encoding (Section 5.4). Six of six is the count; four hard quantitative hits are the evidence.

Recomputed here, on the archive as it stands. The prediction table is the historical record, and later data cannot change it. Separately, our script re-runs the source analysis's

Table 11: Predictions stated before FRB 20220912A was analysed, and their outcomes. This is the strongest methodological feature of the work: the model could not have been tuned to data it had not seen.

prediction	predicted	observed	
Brevity ρ	-0.94 ± 0.05	-0.883	confirmed (within 2σ)
Zipf slope	-0.8 to -1.1	-1.057	confirmed
Same 4-sector phase structure	yes	yes	confirmed
All spectral channels above null	yes	$z = 196$–338	confirmed
Sinusoidal 32-offset fit	$r > 0.5$	$r = 0.595$, $p = 0.0003$	confirmed
Anti-synchronised transitions	yes	yes	confirmed
Vocabulary growth β	0.373 ± 0.02	0.523 (spectral layer)	no object at source

pipeline for this source on the 949 spectra now public; the original run used the 894 then available [28]. The architecture replicates: a dominant carrier (76% of variance) wrapping components that all carry sequential structure at $z = 201$–335 (original run: 196–338), and the 32-offset wave profile persists at the same period ($r = 0.496$, $p = 0.004$). The linguistic values on the grown archive: Zipf slope -1.143, vocabulary growth 0.540, brevity -0.850. We print these as they fall: on today's larger archive the wave-fit point estimate sits below the pre-registered $r > 0.5$ threshold while remaining significant at the same period, and the Zipf slope steepens slightly past the predicted bracket. The carrier, the channel structure and the frame profile are the replicated architecture.

Retrospective cross-source agreement. Separately, ten protocol and statistical features were compared across all three sources after the fact: **25 of 25 testable cells confirmed, none failed**; five cells are not testable from the available data (Fig. 8). This is weaker evidence than the prospective table, because the features were not committed to in advance, and we label it as such.

Prediction confirmed on data not used to build the model is worth more than any single significance value in this paper.

6.1 The same source through two telescopes: a fingerprint square

Every payload measurement so far, on all three sources, comes through one instrument chain: FAST. The remaining objection class is instrumental, that the payload statistics are a signature of the telescope rather than of the sources. This is testable with public data, and we tested it with the match criteria fixed before anything was computed. FRB 121102 is the one source in this paper's set with published burst spectra from a second major telescope: 404 burst-centred, bandpass-normalised Arecibo arrays from the 2016 burst storm [7], three years before the FAST campaign [15]. That yields a two-by-two design: the same source through two telescopes, and different sources through the same telescope.

All four datasets, FRB 121102 through Arecibo, FRB 121102 through FAST, FRB 20201124A and FRB 20220912A through FAST, were reduced to the identical representation: per-burst spectra on a common 12.5-MHz grid over 1150–1500 MHz, the band the two telescopes share, then the identical pipeline used everywhere in this paper (PCA, carrier discarded, the two quadrant streams), then nine stream statistics per 352-burst window, each invariant under relabeling of the quadrants so that no dataset's arbitrary PCA orientation can matter. The distance between two datasets is the median feature difference in units of the within-dataset window scatter.

Table 12 is the result. The same source through two different telescopes: distance 1.27, and **0.70, below the window scatter itself**, under a uniform burst-window extraction of the FAST cell. FRB 121102 against different sources through its own telescope: 3.78 and 5.47 (4.94

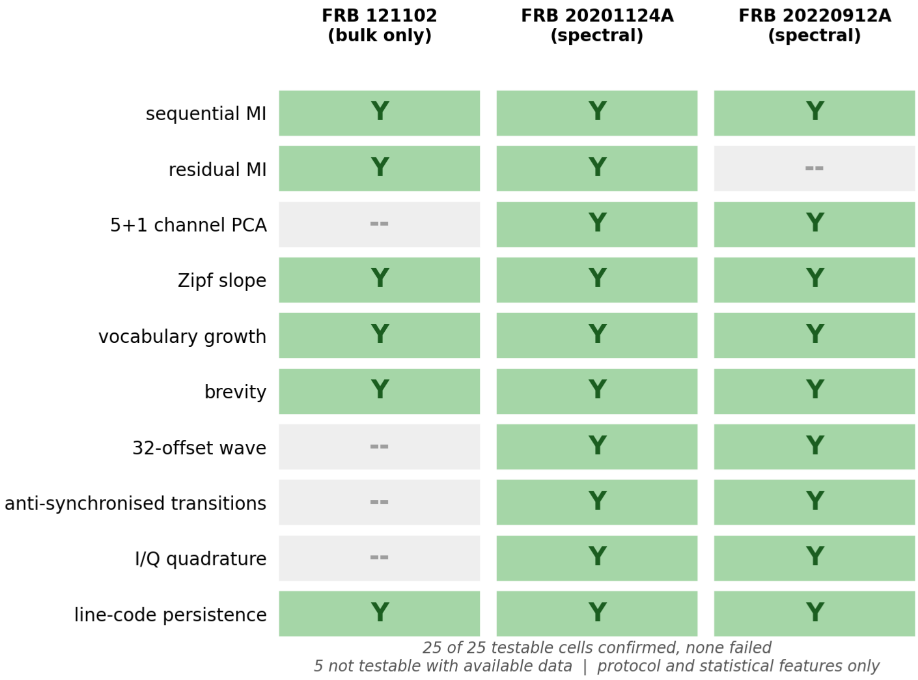

Figure 8: Retrospective feature confirmation across three independent FRB sources, restricted to protocol and statistical features; content-interpretation rows from the source analysis [18] are excluded as outside this paper's scope. Green confirmed, grey not testable with the available data for that source. No cell failed. Compare Table 11, which is prospective and carries more weight.

Table 12: Fingerprint distances between the four datasets: median difference over nine relabeling-invariant payload-stream statistics, in units of within-dataset window scatter. The same source through two different telescopes is three to four times closer than FRB 121102 against either different source through its own telescope (3.78, 5.47); under the uniform burst-window extraction of the FAST 121102 cell the same-source distance falls to **0.70**, below window scatter, against 4.94 and 6.63. The remaining FAST pair separates at 1.65: every pair of different sources is resolved, and the closest pair in the matrix is the same source through two telescopes. All values printed by `fingerprint_square.py`.

	121102/AO	121102/FAST	20201124A/FAST	20220912A/FAST
121102 / Arecibo	—	**1.27**	5.64	7.33
121102 / FAST	**1.27**	—	3.78	5.47
20201124A / FAST	5.64	3.78	—	1.65
20220912A / FAST	7.33	5.47	1.65	—

and 6.63 under the same uniform extraction). The third same-telescope pair, FRB 20201124A against FRB 20220912A, separates at 1.65; every pair of genuinely different sources is resolved, and the closest pair in the whole matrix is the same source through two different telescopes. The control also inverts correctly: if the burst signal is deliberately erased from the FAST 121102 dataset, by averaging whole files so that a millisecond burst is diluted over seconds of receiver noise, the dataset stops matching Arecibo 121102 and drifts toward the other FAST datasets. The telescope's own signature appears exactly and only when the source's signal is removed.

The payload statistics travel with the source, across telescopes and across a three-year epoch gap, while genuinely differing between sources observed through the same instrument; they are modulation-side quantities, not receiver-side ones. An instrumental account of the payload structure would now need FAST to imprint a different fingerprint on each source it observes while Arecibo independently imprints the right one on the one source they share. The caveats are recorded in the companion script's committed output: the two 121102 datasets supply one 352-burst window each, so the scatter scale is taken from the larger datasets, and the channel-order convention is verified against the PSRFITS frequency table for FRB 20201124A and assumed for the other FAST products.

7 Discussion

7.1 What the comparison-class literature actually shows

Sproat [26] compared written language against weather-forecast icon sequences, emoticons, mathematical notation, heraldry, totem poles and Mesopotamian deity symbols. He was asking whether statistics distinguish *writing*, symbols encoding units of a spoken language, from other symbol systems, and he explicitly contrasts writing with systems such as traffic signs, music and mathematics that convey information independently of any language.

His comparison class is therefore **not a set of noncommunicative controls; it is mostly a set of combinatorial meaning-bearing systems**. Emoticons are the clean case: no individual punctuation mark carries the expression the combination produces. Mathematics is unambiguous, since the relations among symbols *are* the proposition. So the result is not that these statistics are empty. It is that they identify a *class* of which written language is one member. Membership is a substantive finding.

Four points bound what the study established. Its classifier contained **no Zipf feature**, so Zipf's known weakness was never bundled with conditional entropy. Its conditional-entropy curve holds conditioning at one symbol and expands the *vocabulary*, making it a vocabulary-growth curve rather than h_1, h_2, h_3; and its block entropies test *levels* $B(k)$, whereas depth lives in the *increments* $h_k = B(k{+}1) - B(k)$, which two corpora can differ in while agreeing on the sum. Its inscriptions averaged 1.4 to 13.6 symbols against 6-symbol blocks. **Depth was never measured.** And its conclusion was not that classification is impossible: a modified cut-off and a new repetition measure classified the categories successfully. Section 5.3 runs its decisive control directly and passes at $z = 48$–54 with zero exceedances.

This framing is the published standard. *Science* carries "Whale song shows language-like statistical structure" [2] and *Nature Communications* carries "Contextual and combinatorial structure in sperm whale vocalisations" [24]. Both are post-Sproat, both in top-tier venues, and both claim exactly what we claim: combinatorial statistical structure, not natural language. We hold ourselves to that standard and meet it with larger margins.

7.2 The direction of the excess

Where the payload deviates from written English, it deviates consistently in one direction: more redundancy (44.2% against matched English's 1.6%), a longer dependency span (18 against 2.3), brevity holding more tightly (-0.921 against -0.70). A natural question is whether the direction itself says something about the *kind* of content, since noise does not overshoot language, but dense, formal, structured text might. We measured it. Under the identical pipeline, mathematical prose (Laplace's *Essay on Probabilities*) sits at $1.9 \pm 0.5\%$ order-1 redundancy with depth 2.9 ± 1.4; structured problem-and-solution text (Dudeney's *Amusements in Mathematics*) at $2.5 \pm 0.7\%$ with depth 2.6 ± 0.8. **Order-1 redundancy climbs monotonically as English text becomes more formal, dependency span stays confined near 2–3 throughout, and neither comes anywhere near the payload.** The gradient runs narrative prose → mathematical prose → structured mathematics → this signal, with the signal roughly an order of magnitude beyond the structured end. The direction is consistent with content more formal than any natural-language register; the magnitude says no English text approaches it. We record this as an observation about the statistics, not a reading of content: what kind of source sits that far along the axis that formal text starts down belongs to the experiments of Section 7.5.

7.3 A note on non-separable objections

Three of the most natural objections to this analysis share a structure worth naming, because each is persuasive on first reading and none of them bites.

1. *The low entropy at offsets 8–16 is caused by the skewed symbol marginal there, not by a header.* But a symbol marginal that varies with offset is what a frame is. The objection relocates the claim from one statistic to another and leaves it intact (Section 2.5). The two quantities involved, mean wave-state amplitude and per-offset entropy, are moreover both summary statistics of the *same* per-offset symbol distribution, so their high correlation is close to an identity rather than a confound.
2. *The Markov surrogate reproduces the redundancy, so the redundancy is just persistence.* But the surrogate was built from the measured transition matrix, and order-1 redundancy is a deterministic function of it. The match is arithmetic (Section 5.4).
3. *The entropy segmentation disagrees with the mutual-information boundaries.* But coupling seams and low-variability fields are different features and sit at different offsets in any framed format (Section 2.2).

The common form is an alternative explanation that is not *separable* from the claim it is meant to replace. A genuine alternative must be capable of being true while the claim is false. If asserting it requires asserting the claim, if it appeals to a quantity that was fixed by construction, or if it requires two statistics measuring different objects to coincide, it has not competed with the claim at all. Stated carefully, each of the three concedes the thing it was raised to deny.

We state this not to deflect scrutiny but to direct it. The separable objections are the ones we have tried hardest to answer, and they are the ones in Section 5: does temporal order matter, is the vocabulary distribution reproducible from persistence alone, is an absolute entropy value reproducible from a skewed marginal, and can a calibrated physical process account for the sequential structure. Those tests can all come out the other way, and two of them nearly do: stream 2's Zipf slope is approached by the Markov surrogate ($z = -1.7$, 10 of 300), and the Ornstein–Uhlenbeck null clears only $z = 2.02$. We report both margins as they fell.

7.4 On "but is it *really* language"

We refuse this question, and we refuse it on principle rather than for want of an answer.

Whether a system that transmits information through rule-governed symbol combinations also earns the word *language* is a question about the boundaries of an English term. It is settled by definition, not by measurement, and **no observation of the signal can bear on it.** That is not a hard question. It is a question of a kind that measurement cannot address at all.

We therefore name what is happening when it is demanded as a precondition. **A criterion that no possible observation can satisfy, required before measurements are taken seriously, is not a high standard. It is a bias toward the null wearing the costume of rigour.** It cannot be met by better data, more surrogates, or a further control, because nothing in the data is what it is about. Applied consistently it would also exclude the whale-song results in *Science* and *Nature Communications* that supply the register for this paper [2, 24], and it is not applied to them. We decline to accept it as a legitimate mode of academic examination of this subject, and we say so plainly rather than negotiating with it.

The measurable questions are these, and they are ordered:

1. Is the sequence nonrandom?
2. **Do combinations carry information not available from the symbols individually?**
3. Is it a conventional meaning-bearing communication system?
4. Does it encode a natural human language?

Within this paper, *language* is accordingly used the way the toolkit uses it: as the name of the measured class that questions 1 and 2 pick out, systems whose symbol combinations carry information their symbols alone do not. Questions 3 and 4 name different properties, and none of them is claimed.

This paper answers 1 and 2 affirmatively, with margins running from $z = 8.4$, against a surrogate built to reproduce the signal's own persistence, **to** $z = 512$ against an order shuffle. Question 2 is the substantive one. A system whose combinations carry information beyond its symbols is doing what mathematics, DNA, music, cetacean song and written language all do, and there is no well-attested natural system with deep combinatorial symbolic dependency that is not information-bearing.

Questions 3 and 4 require different evidence, and we say what would supply it: conditioning one source's output on another's preceding sequence, directional transfer entropy between them, cross-source replication of the same field layout, and systematic covariance between symbol combinations and an independently measurable variable. Those are experiments, and they are the next ones.

7.5 Further protocol machinery available

We applied frame-period estimation with falsification against shorter periods, per-offset entropy field segmentation, line-code persistence and transition-grammar asymmetry. The standard battery contains more, and each is a further test this signal can be put to: randomness testing of the content stream against structured-header controls, sequence alignment across frames, explicit length-field inference, and preamble detection by cross-correlation. Each has a definite expected outcome under the protocol reading, which is the useful property: this reading is not finished being tested, and it is not difficult to test further. The conditional transfer-entropy test of Section 2.4 belongs to the same battery.

7.6 What was done, and what it means

The argument of this paper is short enough to state in four sentences.

We took the standard toolkit of automatic protocol reverse engineering, the one that is checked daily against protocols whose specifications are known, and applied it unmodified to a burst sequence. It found a transport layer: a pre-specified 32-symbol frame that falsifies both of its sub-period alternatives, a carrier wrapping five sub-channels confirmed without any decomposition, and a line code holding state two and a half times more often than its own memoryless rate. We then performed the operation those tools exist to enable, which is the operation a receiver performs: discard the carrier and demultiplex the sub-channels.

We took the standard toolkit of statistical language detection, established in the animal-communication and SETI literature and carrying the claim register of the recent cetacean results in *Science* and *Nature Communications*, and applied it unmodified to what came out. It passed on both axes: on the primary stream, a Zipf slope of -0.890 whose maximum-likelihood form the strict goodness-of-fit protocol does not reject, vocabulary growth of 0.507 against 0.50, and a sequential-dependency depth of 18 against matched-pipeline English at 2.3 ± 0.8; on the convergent second stream, a Zipf slope of -1.057 against English's -1.00 and brevity exceeding English's. Against a surrogate matched on persistence and run lengths by construction, the incremental contribution of second-order context exceeds the null at $z = +9.4$ and $+8.4$.

We detected a protocol. We stripped it. We detected language. By the standards of the two fields whose tools we used, and using no method either field would not recognise, these are excellent signals for exactly those two things.

Code and data availability

FRB 20201124A, 1,863 bursts, FAST [27]; dynamic spectra and the burst-parameter table from doi:10.57760/sciencedb.j00113.00076. FRB 121102, 1,652 bursts, FAST [15]; burst catalogue from CDS (`J/Nature/598/267/tables1.dat`). FRB 20220912A, FAST [28]; dynamic spectra from doi:10.57760/sciencedb.08058, 949 public at this writing. Each burst contributes its time-averaged dynamic spectrum over 512 frequency channels. PCA on the (bursts $\times$ channels) matrix; PC1 (84% of variance, plain brightness) discarded. The next four components form two channel pairs, each giving one angle per burst, quantised into four quadrants. Stream 1 is the first pair; stream 2 is the first pair relative to the second.

Every number computed for the core three-part reanalysis is printed by `python frb_three_part.py`, a single script requiring only `numpy`, `scipy` and `pandas`, run over seven public data files (`spectra.npy`, `tables1.dat.txt`, `frb20201124a_parameters.csv`, `spectra_20220912a.npy`, and three public-domain control corpora from Project Gutenberg: `english_sample.txt`, ebook #2701; `math_sample_prose.txt`, #58881; `math_sample_puzzles.txt`, #16713). The script is deterministic: every stochastic block carries its own seed, so repeated runs are byte-identical and the values are exactly reproducible. `build_spectra.py` and `build_spectra_20220912a.py` rebuild the two spectral matrices from the archive downloads. Two companion scripts follow the same discipline and ship with their committed outputs: `fingerprint_square.py` (the cross-telescope square of Section 6.1; Arecibo burst arrays from doi:10.5281/zenodo.7181266 [7], FAST 121102 sub-integration spectra rebuilt from the FAST-FREX archive doi:10.57760/sciencedb.15070 by `build_spectra_frb121102_fast.py`) and `depth_gain_20240114a.py` (the large-N bulk-parameter test of Section 3; burst table from doi:10.57760/sciencedb.Fastro.00030 [29]). The handful of results quoted from the source analysis rather than recomputed here, principally the Ornstein–Uhlenbeck and residual tests, the directed cross-burst dependency, and the pre-registered FRB 20220912A prediction outcomes, are cited to it where they appear. No trained model, no GPU, no non-standard dependency. This is a condensed reanalysis of [18], which contains the full multi-source treatment.

References

[1] Aggarwal, K., Agarwal, D., Lewis, E. F., et al. (2021). Comprehensive analysis of a dense sample of bursts from the repeating fast radio burst FRB 121102A. *Astrophys. J.* **922**, 115.

[2] Arnon, I., et al. (2025). Whale song shows language-like statistical structure. *Science*. doi:10.1126/science.adq7055

[3] Clauset, A., Shalizi, C. R., & Newman, M. E. J. (2009). Power-law distributions in empirical data. *SIAM Review* **51**(4), 661–703.

[4] Corral, Á., Serra, I., & Ferrer-i-Cancho, R. (2020). Distinct flavors of Zipf's law and its maximum likelihood fitting. *Phys. Rev. E* **102**, 052113.

[5] De Gregorio, J., Sánchez, D., & Toral, R. (2026). Information-theoretic analysis of temporal dependence in discrete stochastic processes: application to precipitation predictability. *Chaos* **36**(3), 033124.

[6] Ferrer-i-Cancho, R., & McCowan, B. (2012). The span of correlations in dolphin whistle sequences. *J. Stat. Mech.* 2012, P06002.

[7] Hewitt, D. M., Snelders, M. P., Hessels, J. W. T., et al. (2022). Arecibo observations of a burst storm from FRB 20121102A in 2016. *Mon. Not. R. Astron. Soc.* **515**(3), 3577–3596.

[8] Bossert, G., Guihéry, F., & Hiet, G. (2014). Towards automated protocol reverse engineering using semantic information. *Proc. 9th ACM Symposium on Information, Computer and Communications Security (ASIACCS)*, 51–62. (Netzob.)

[9] Chandler, J., Wick, A., & Fisher, K. (2023). BinaryInferno: a semantic-driven approach to field inference for binary message formats. *Network and Distributed System Security Symposium (NDSS) 2023.*

[10] Duchêne, J., Le Guernic, C., Alata, E., Nicomette, V., & Kaâniche, M. (2018). State of the art of network protocol reverse engineering tools. *Journal of Computer Virology and Hacking Techniques* **14**, 53–68.

[11] Kleber, S., Maile, L., & Kargl, F. (2019). Survey of protocol reverse engineering algorithms: decomposition of tools for static traffic analysis. *IEEE Communications Surveys & Tutorials* **21**(1), 526–561.

[12] Proakis, J. G., & Salehi, M. (2008). *Digital Communications* (5th ed.). McGraw-Hill.

[13] Dorfinger, P., Panholzer, G., & John, W. (2011). Entropy estimation for real-time encrypted traffic identification. *Traffic Monitoring and Analysis*, 164–171.

[14] Doyle, L. R., McCowan, B., Johnston, S., & Hanser, S. F. (2011). Information theory, animal communication, and the search for extraterrestrial intelligence. *Acta Astronautica* **68**, 406–417.

[15] Li, D., Wang, P., Zhu, W. W., et al. (2021). A bimodal burst energy distribution of a repeating fast radio burst source. *Nature* **598**, 267–271.

[16] Lyda, R., & Hamrock, J. (2007). Using entropy analysis to find encrypted and packed malware. *IEEE Security & Privacy* **5**(2), 40–45.

[17] McCowan, B., Hanser, S. F., & Doyle, L. R. (1999). Quantitative tools for comparing animal communication systems. *Animal Behaviour* **57**, 409–419.

[18] Meucci, S. (2026). Evidence for structured signal traffic in fast radio burst data. doi:10.5281/zenodo.19443570

[19] Moreno-Sánchez, I., Font-Clos, F., & Corral, Á. (2016). Large-scale analysis of Zipf's law in English texts. *PLoS ONE* **11**(1), e0147073.

[20] Paninski, L. (2003). Estimation of entropy and mutual information. *Neural Computation* **15**(6), 1191–1253.

[21] Rajabi, F., Houde, M., Bartkiewicz, A., et al. (2020). A simple relationship for the spectro-temporal structure of bursts from FRB 121102. *MNRAS* **498**, 4936–4942.

[22] Schreiber, T. (2000). Measuring information transfer. *Phys. Rev. Lett.* **85**, 461–464.

[23] Shannon, C. E. (1951). Prediction and entropy of printed English. *Bell Syst. Tech. J.* **30**, 50–64.

[24] Sharma, P., et al. (2024). Contextual and combinatorial structure in sperm whale vocalisations. *Nature Communications* **15**, 3617. doi:10.1038/s41467-024-47221-8

[25] Sija, B. D., Goo, Y.-H., Shim, K.-S., Hasanova, H., & Kim, M.-S. (2018). A survey of automatic protocol reverse engineering approaches, methods, and tools. *Security and Communication Networks* **2018**, 8370341.

[26] Sproat, R. (2014). A statistical comparison of written language and nonlinguistic symbol systems. *Language* **90**(2), 457–481.

[27] Wang, P., Li, D., Feng, Y., et al. (2023). Atlas of dynamic spectra of fast radio burst FRB 20201124A. *Chinese Physics B* **32**, 029801.

[28] Zhang, Y.-K., Li, D., Zhang, B., et al. (2023). FAST observations of FRB 20220912A: burst properties and polarization characteristics. *Astrophys. J.* **955**, 142.

[29] Zhang, J.-S., et al. (2025). The magnetar model's energy crisis for a prolific repeating fast radio burst source. arXiv:2507.14707. Burst table: doi:10.57760/sciencedb.Fastro.00030.

[30] Zipf, G. K. (1949). *Human Behavior and the Principle of Least Effort.* Addison-Wesley.